

Evidence Against Equilibrium: Optimal Rates and Anytime-Valid Tests of Detailed Balance

Joshua Winters
Independent Researcher
josh@friendmachine.co

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Abstract

A stationary Markov chain that breaks detailed balance is out of equilibrium, and its trajectories look different when played backward. We study how to test for broken detailed balance while the data arrive. A likelihood-ratio test calibrated to 5% at a planned sample size loses that guarantee when it is checked repeatedly; in our simulations its false-alarm rate rose to 24–60%. We give tests that remain valid when checked after every observation, and we identify the rate that limits any such test. Against the composite null hypothesis of reversibility, the expected evidence of a valid test grows at most at rate $\rho = \sum_{ij} f_{ij} \log_2 \frac{2f_{ij}}{f_{ij}+f_{ji}}$ bits per step, where f_{ij} is the stationary edge flow. This rate is the divergence to the nearest reversible chain, not the entropy production σ : $\rho \leq \sigma/2$, $\rho \approx \sigma/4$ near equilibrium, and ρ stays finite when σ is infinite. Together with a recent general lower bound for sequential testing with Markov data, this implies an expected detection time of at least about $\log_2(1/\alpha)/\rho$ steps at level α . A likelihood e-process built by universal inference reaches rate ρ asymptotically but is slow at practical α . Our main proposal, the loop test, bets on the orientation of loops through a reference state. It is valid at every sample size under continuous monitoring and needs no optimization. For $\rho \leq 0.02$ bits/step it captures 92–100% of the optimal rate; for $\rho < 0.07$ it captures 80–100% and needs 1.7–1.9 times the sample size of an idealized fixed-sample test.

1 Introduction

A stationary Markov chain with transition matrix T and stationary law π is *reversible*, or satisfies detailed balance, if $\pi_i T_{ij} = \pi_j T_{ji}$ for all states i, j . Detailed balance holds at thermodynamic equilibrium. Its violation shows that the system is driven, for example by ATP hydrolysis in molecular motors or by external forcing in active matter. The entropy production rate σ measures the violation, and many methods estimate it from observed trajectories (Section 2).

We study a question that comes before estimation: how to decide, while data arrive, whether detailed balance is broken. A recording is rarely analysed once at a length fixed in advance. Experimenters look as the data accumulate and stop when the evidence seems convincing, and classical tests lose their guarantees under this practice. In our simulations of reversible chains (Section 7), a likelihood-ratio test calibrated to 5% at a fixed sample size raised false alarms in 23.9–35.5% of runs when checked 10 times, and in 44.5–59.8% when checked at 221 points along the trajectory.

Section 4 identifies the rate that governs this decision. Against the composite null “some reversible chain,” the expected log-evidence of any valid test grows at most at rate ρ , the Kullback–Leibler rate from T to its nearest reversible chain $(T + T^*)/2$. We show that $\rho \leq \sigma/2$, that $\rho \approx \sigma/4$

near equilibrium, and that ρ is finite even when σ is infinite. The nearest reversible chain is known [25, 1], and the resulting bound on detection time, $\mathbb{E}[\tau] \gtrsim \log_2(1/\alpha)/\rho$, is a special case of a general result of Sethi et al. [35]. What is new is the closed form of the rate for this null and its relation to σ .

Sections 5 and 6 give two tests. The first is an e-process built by running maximum likelihood [24]. Its evidence grows at exactly rate ρ , but it pays a model-selection cost of order $(k(k-1)/2) \log t$ bits, which dominates at practical significance levels. The generalized-likelihood test of [35], applied to this null, is a close relative with the same cost. The second, the loop test, is our main methodological contribution. Under detailed balance, each loop through a reference state is as likely as its reversal, so the loop’s orientation is a fair coin. Betting on orientations gives an anytime-valid test, one that may be checked after every observation, and it needs no constrained optimization. Its overhead is small: for $\rho \leq 0.02$ bits/step it captures 92–100% of the optimal rate.

Section 7 checks these claims in simulation. Both tests hold their level under repeated monitoring, and the loop test does so when checked after every step. The evidence grows at the predicted rates. For $\rho < 0.07$, the loop test detects broken detailed balance within 1.7–1.9 times the sample size of an idealized fixed-sample test.

This paper supersedes an earlier version organized around “evidence holonomy,” which described irreversibility as the change in description length around a loop of representation transforms. The entropy-production estimator from that version is in Appendix A, and the framing is summarized in Appendix B.

2 Related work

Entropy production and time reversal. For stationary Markov processes, the relative-entropy rate between forward and time-reversed path measures equals the entropy production rate [20, 6, 21]. Roldán and Parrondo estimated this divergence from single trajectories with plug-in and compression-based estimators, the latter via Ziv–Merhav cross-parsing [18, 19, 26]. Chazottes and Redig [2] and Cristadoro et al. [3] built universal estimators from hitting and waiting times. Coarse-grained or partial observation yields lower bounds on dissipation [7], and waiting-time statistics can reveal irreversibility that transition counts miss [12]. Compression [13, 16] and neural networks [10] have also been used to estimate entropy production. Cycle counts and affinities in Markov models are analyzed in [14]. In biophysics, broken detailed balance is detected from probability-flux loops in coarse-grained phase space, as for beating flagella [27]; see the review [30] and an application to brain dynamics [33]. Time-series irreversibility tests are reviewed in [36].

Deciding the arrow of time. Roldán et al. [17] apply Wald’s sequential probability ratio test to decide the direction of time’s arrow. Their test compares a process with its own time reversal, a *simple* null, and evidence accrues at rate σ . We test the *composite* null “some reversible chain,” which asks whether the system is at equilibrium at all, without specifying which equilibrium. Section 4 shows the relevant rate is then $\rho \leq \sigma/2$.

Testing reversibility. Fixed-sample tests of reversibility for Markov chain data include chi-squared tests on transition counts [22] and algebraic checks [9]. Diaconis and Rolles give a conjugate prior for reversible chains [28], which would allow a Bayesian mixture in place of the maximum likelihood used in Section 5. Property testing studies identity testing of reversible chains [5]. Kolmogorov’s criterion characterizes reversibility by cycle products [31]. The reversible chains form both an exponential and a mixture family; the m-projection of a kernel onto them is the

additive reversibilization $(T + T^*)/2$ [25, 4], identified as the closest equilibrium dynamics in [1]. Our Lemma 4.1 restates this result.

Sequential and anytime-valid testing. E-values and e-processes give tests that stay valid under optional stopping and continuation [8, 15]. Universal inference constructs them for composite nulls from a predictable likelihood divided by the null maximum likelihood [24]. For nulls defined by invariance under a group, optimal e-statistics are likelihood ratios of maximal invariants [34, 32]. The loop test of Section 6 is a betting test of such a symmetry: the orientation flip of each loop. Most directly related, Sethi et al. [35] study sequential tests of a composite null set of transition matrices from Markov data. They prove an instance-dependent lower bound on the expected stopping time, $\log(1/\alpha)$ divided by the infimum over the null of a stationary-weighted KL divergence, and give a generalized-likelihood test that attains it as $\alpha \rightarrow 0$. They do not treat reversibility. For the reversible null, Lemma 4.1 evaluates their infimum in closed form as ρ . Our Theorem 4.3(b) is therefore a special case of their Theorem 3.3, and their test, instantiated for this null, is a rate-optimal alternative to our Section 5. We found no prior test of detailed balance based on loop orientations, and no prior comparison of the optimal composite-null rate with σ .

3 Setup

Let $x_0, x_1, \dots$ be a Markov chain on $\mathcal{X} = \{1, \dots, k\}$ with transition matrix T . When T is irreducible, π denotes its stationary law, $f_{ij} = \pi_i T_{ij}$ the stationary edge flow, and $T_{ij}^* = \pi_j T_{ji} / \pi_i$ the time reversal. The entropy production rate is

$$\sigma = \sum_{i,j} f_{ij} \log_2 \frac{f_{ij}}{f_{ji}} \quad (\text{bits/step}),$$

equal to $+\infty$ if some transition is possible in only one direction [6]. We call T *reversible* if it satisfies Kolmogorov's criterion [31]: for every closed walk $z_0, z_1, \dots, z_m = z_0$,

$$\prod_{s=1}^m T_{z_{s-1} z_s} = \prod_{s=1}^m T_{z_s z_{s-1}}.$$

For irreducible T this is equivalent to detailed balance $\pi_i T_{ij} = \pi_j T_{ji}$; the criterion also covers reducible chains, which may arise as maximizers of the likelihood at finite t . The null hypothesis is

$\mathcal{H}_0$: the data are a Markov chain with some reversible T , from any initial state.

Results under the alternative (Sections 4–5) assume T irreducible; aperiodicity is not needed. A *level- α anytime-valid test* is a process (E_t) , adapted to the data, such that $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$ for every law in $\mathcal{H}_0$. It may be checked after every observation. An *e-process* satisfies the stronger condition $\mathbb{E}[E_\tau] \leq 1$ for every stopping time τ and every null law. Every nonnegative martingale with initial value 1 is an e-process, and Ville's inequality makes it a level- α test. Logarithms are base 2 unless stated.

4 The optimal rate

Lemma 4.1 (Nearest reversible chain [25]). *Let T be irreducible. Among reversible transition matrices Q , the rate $D_\pi(T \| Q) = \sum_{i,j} f_{ij} \log_2(T_{ij}/Q_{ij})$ is minimized by the additive reversibilization*

$M = (T + T^*)/2$, and

$$\min_{Q \text{ reversible}} D_\pi(T\|Q) = \rho := \sum_{i,j} f_{ij} \log_2 \frac{2f_{ij}}{f_{ij} + f_{ji}}.$$

Proof. Let g be the symmetric edge measure of a reversible Q , with marginal μ . Then $D_\pi(T\|Q) = D(f\|g) - D(\pi\|\mu)$. Let $s = (f + f^\top)/2$, which is symmetric with marginal π by stationarity. Since $\log(s/g)$ is symmetric, $\sum f \log(s/g) = \sum s \log(s/g)$, hence $D(f\|g) = D(f\|s) + D(s\|g)$. By data processing, $D(s\|g) \geq D(\pi\|\mu)$. So $D_\pi(T\|Q) \geq D(f\|s) = \rho$, with equality when $g = s$, that is, $Q = M$. $\square$

Proposition 4.2 (Properties of ρ). (a) $0 \leq \rho \leq \sigma/2$, and $\rho = 0$ if and only if T is reversible.

(b) $\rho \leq \sum_{i \neq j} f_{ij}$ bits. In particular, $\rho < \infty$ even when $\sigma = \infty$.

(c) If $f_{ij} = s_{ij} + a_{ij}$ and $f_{ji} = s_{ij} - a_{ij}$ with $|a_{ij}| \ll s_{ij}$, then $\rho = \sigma/4 + O(\max |a|^4/s^3)$.

Proof. (a) By convexity of D in its second argument, $D(f\|\frac{1}{2}f + \frac{1}{2}f^\top) \leq \frac{1}{2}D(f\|f) + \frac{1}{2}D(f\|f^\top) = \sigma/2$. (b) Each term satisfies $\log_2(2f_{ij}/(f_{ij} + f_{ji})) \leq 1$, and diagonal terms vanish. (c) For each unordered pair, ρ contributes $(s + a) \log \frac{s+a}{s} + (s - a) \log \frac{s-a}{s} = a^2/s + a^4/(6s^3) + O(a^6/s^5)$ nats, while σ contributes $2a \log \frac{s+a}{s-a} = 4a^2/s + 4a^4/(3s^3) + O(a^6/s^5)$ nats. $\square$

Theorem 4.3 (No test can do better). Let T be irreducible and irreversible, and let P denote the law of the chain started from π .

- (a) Let (E_t) be any nonnegative adapted process with $\mathbb{E}_Q[E_t] \leq 1$ for every fixed t and every $Q \in \mathcal{H}_0$; every e -process qualifies. Then $\mathbb{E}_P[\log_2 E_t] \leq t\rho$ for all t .
- (b) Consider a sequential test that stops at a time τ with $\mathbb{E}_P[\tau] < \infty$ and then either rejects or accepts $\mathcal{H}_0$. Suppose its probability of rejecting is at most α under every law in $\mathcal{H}_0$ and at least $1 - \beta$ under P , with $\alpha + \beta < 1$. Then

$$\mathbb{E}_P[\tau] \geq \frac{d(1 - \beta \parallel \alpha) - c_T}{\rho},$$

where $d(p\|q) = p \log_2 \frac{p}{q} + (1 - p) \log_2 \frac{1-p}{1-q}$, and $c_T = \max h - \min h$ is the oscillation of the solution h of the Poisson equation $(I - T)h = \bar{g} - \rho \mathbf{1}$ defined in the proof; it depends only on T . As $\alpha, \beta \rightarrow 0$, the bound is $(1 + o(1)) \log_2(1/\alpha)/\rho$.

Part (b) is the reversible-null case of the general lower bound of Sethi et al. [35, Thm. 3.3], whose denominator, the infimum over the null of a stationary-weighted KL divergence, equals ρ by Lemma 4.1. We include a self-contained proof.

Proof. Let $M = (T + T^*)/2$, and let P^M be the law of the chain with transition matrix M started from π . M is reversible with stationary law π , so $P^M \in \mathcal{H}_0$. Every transition possible under T is possible under M , because $M_{ij} \geq T_{ij}/2$. Hence $P_t \ll P_t^M$. The per-step log-likelihood ratio $g(i, j) = \log_2(T_{ij}/M_{ij})$ is bounded on the finitely many edges with $T_{ij} > 0$, and $g \leq 1$. Its conditional mean $\bar{g}(i) = \sum_j T_{ij}g(i, j)$ satisfies $\sum_i \pi_i \bar{g}(i) = \rho$ by Lemma 4.1. Both chains start from π , so $D(P_t\|P_t^M) = t\rho$ exactly. From another initial law used by both chains, it is $t\rho + O(1)$.

- (a) By Gibbs' inequality, $\mathbb{E}_P[\log_2 E_t] \leq D(P_t\|P_t^M) + \log_2 \mathbb{E}_{P^M}[E_t] \leq t\rho$.

- (b) Let $L_\tau = \sum_{s \leq \tau} g(x_{s-1}, x_s)$. By data processing applied to the rejection event, $\mathbb{E}_P[L_\tau] = D(P_\tau \| P_\tau^M) \geq d(P(\text{reject}) \| P^M(\text{reject})) \geq d(1 - \beta \| \alpha)$. The last step uses $P(\text{reject}) \geq 1 - \beta > \alpha \geq P^M(\text{reject})$ and the monotonicity of $d(p \| q)$ in each argument when $p > q$. Since T is irreducible on a finite space, the Poisson equation $(I - T)h = \bar{g} - \rho \mathbf{1}$ has a bounded solution h . Writing

$$L_\tau - \rho\tau = \sum_{s \leq \tau} [g(x_{s-1}, x_s) - \bar{g}(x_{s-1})] + \sum_{s \leq \tau} [h(x_s) - (Th)(x_{s-1})] + h(x_0) - h(x_\tau)$$

expresses $L_\tau - \rho\tau$ as a martingale with bounded increments plus a term bounded by c_T . Optional stopping applies because $\mathbb{E}_P[\tau] < \infty$ [29]. It gives $\mathbb{E}_P[L_\tau] \leq \rho \mathbb{E}_P[\tau] + c_T$. Combining the two bounds proves the claim. For the limit, $d(1 - \beta \| \alpha) = (1 + o(1)) \log_2(1/\alpha)$ as $\alpha, \beta \rightarrow 0$. $\square$

Remark 4.4 (Why ρ and not σ). Roldán et al. [17] test the direction of time's arrow against a simple null, the chain's own time reversal P^* , and their evidence accrues at rate $D(P \| P^*) = \sigma$. Our null is composite. Its least favorable member is M , not P^* , so evidence accrues at rate $\lim_t \frac{1}{t} D(P_t \| P_t^M) = \rho \leq \sigma/2$. Asking whether a system is at equilibrium at all, without specifying which equilibrium, is harder than asking which way time runs.

5 A rate-optimal likelihood e-process

Let $C_{ij}(t)$ count the transitions $i \rightarrow j$ among the first t , and let $q_{s-1}(j | i) = (C_{ij}(s-1) + \frac{1}{2}) / (\sum_l C_{il}(s-1) + \frac{k}{2})$ be the Krichevsky–Trofimov (KT) predictor [11]. Define

$$E_t = \frac{\prod_{s=1}^t q_{s-1}(x_s | x_{s-1})}{\sup_{Q \text{ reversible}} \prod_{s=1}^t Q(x_{s-1}, x_s)}.$$

The denominator is the reversible maximum likelihood, over all Q satisfying Kolmogorov's criterion, including reducible ones. We compute it with the standard self-consistent iteration $X_{ij} \leftarrow (C_{ij} + C_{ji}) / (N_i/x_i + N_j/x_j)$ on a symmetric matrix X , where $N_i = \sum_j C_{ij}$ and $x_i = \sum_j X_{ij}$; the maximizer is $Q_{ij} = X_{ij}/x_i$ [23].

Proposition 5.1 (Validity). *Under $\mathcal{H}_0$, (E_t) is an e-process. In particular, $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$ for any initial state.*

Proof. If T_0 is the true reversible matrix, the denominator is at least $\prod_s T_0(x_{s-1}, x_s)$. So $E_t \leq \prod_s q_{s-1}(x_s | x_{s-1}) / T_0(x_{s-1}, x_s)$, which is a nonnegative martingale with initial value 1 [24]. $\square$

Proposition 5.2 (Growth). *If T is irreducible, $\frac{1}{t} \log_2 E_t \rightarrow \rho$ almost surely.*

Proof sketch. The KT numerator is within $\frac{k(k-1)}{2} \log_2 t + O_k(1)$ bits of the unconstrained maximum likelihood. The empirical edge frequencies converge to f . The unconstrained and reversible maximized log-likelihoods, divided by t , converge to $\sum f \log T$ and $\max_{Q \text{ rev}} \sum f \log Q$; the second step uses continuity of the reversible maximum in the frequencies. Their difference is ρ by Lemma 4.1. $\square$

By Theorem 4.3(a), (E_t) is growth-rate optimal. At finite t , however, its log-evidence is roughly $\rho t - \frac{k(k-1)}{2} \log_2 t$, plus fluctuations of order $\sqrt{t}$. The numerator pays for predicting every transition probability, while the denominator fits the symmetric part for free. At $\alpha = 0.05$ this overhead

dominates. The heuristic detection time, the solution of $\rho\tau = \log_2(1/\alpha) + \frac{k(k-1)}{2} \log_2 \tau$, overpredicts the observed medians by a factor of 1.2–1.9, but it identifies the source of the cost.

Numerical note. The iteration is stopped by a relative-change rule, not a certified optimality gap. In a separate check (Section 7) against multi-start quasi-Newton optimization on 150 random count matrices, it fell short of the maximum by at most 1.5×10^{-4} nats, which inflates E_t by a factor of at most about 1.00015.

6 The loop test

Construction. Count time in transitions. Let $B = 10k$ and let r be the most frequent state among $x_0, \dots, x_B$, with ties going to the lowest index. From the first time $t \geq B$ with $x_t = r$, cut the path into *loops* $e = (y_0, y_1, \dots, y_m)$ with $y_0 = y_m = r$ and $y_s \neq r$ for $0 < s < m$. Let $\bar{e}$ denote the reversed loop. When a loop that started at time t_0 closes, multiply E (initially 1) by

$$\frac{2q(e)}{q(e) + q(\bar{e})}, \quad q(e) = \prod_{s=1}^m q(y_s | y_{s-1}), \quad q(\bar{e}) = \prod_{s=1}^m q(y_{s-1} | y_s),$$

where q is the KT estimate from all transitions $1, \dots, t_0$, frozen at the loop's start. Palindromic loops give a factor of exactly 1. Reject reversibility the first time $E \geq 1/\alpha$.

Theorem 6.1 (Validity of the loop test). *Under $\mathcal{H}_0$, $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$.*

Proof. By the strong Markov property, each loop is independent of the past at its start, including the choice of r (which uses only $x_0, \dots, x_B$) and the frozen estimate. For a loop $e = (y_0, \dots, y_m)$, the probability of traversing it from r is $P(e) = \prod_{s=1}^m T_{y_{s-1}y_s}$, and that of its reversal is $P(\bar{e}) = \prod_{s=1}^m T_{y_sy_{s-1}}$. Kolmogorov's criterion makes these equal. Given the unordered pair $\{e, \bar{e}\}$ and the past, the observed orientation is therefore a fair coin, and each factor has conditional mean $\frac{q(e)}{q(e)+q(\bar{e})} + \frac{q(\bar{e})}{q(e)+q(\bar{e})} = 1$. Indexed by completed loops, E is a nonnegative martingale with initial value 1. It is constant between completions, so its supremum over all steps equals its supremum over completions. Ville's inequality gives the claim. If r is never revisited, no loop closes and $E \equiv 1$. $\square$

Remark 6.2 (Fixed times). For each fixed t , $\mathbb{E}_Q[E_t] = 1$ under every $Q \in \mathcal{H}_0$. Given the path up to the start of the first loop and all the unordered loops $\{e, \bar{e}\}$, the set of loops completed by time t is determined, because a loop and its reversal have the same length. The orientations are then independent fair coins, each bet with a predictable estimate. So E_t is a product of mean-one factors over a determined set of loops. Theorem 4.3(a) therefore applies to the loop test.

Remark 6.3 (What is not claimed). (E_t) is not an e-process with respect to all step-level stopping times. A stopping rule that inspects a partially completed loop learns about that loop's orientation and can make $\mathbb{E}[E_\tau]$ exceed 1. The AI-assisted review described in Section 7 found such a rule, with $\mathbb{E}[E_\tau] \approx 1.18$. Fixed times are safe (Remark 6.2). The guarantee is Theorem 6.1: the loop test is an anytime-valid level- α test. For the same reason, averaging the processes for different reference states is not covered, because they evolve on different clocks.

Efficiency. With oracle knowledge of T , the evidence per loop is $\mathbb{E}[\log_2(2P(e)/(P(e) + P(\bar{e})))]$. Divided by the mean loop length, this is at most ρ , by Theorem 4.3(a) and Remark 6.2. Table 1 gives the ratio. For $\rho \leq 0.02$ bits/step, loop orientations carry 92–100% of the available evidence. At $\rho \approx 0.06$ they carry 80%, and far from equilibrium as little as 34%, because the test discards within-loop information. Chains that far from equilibrium are still detected within about a hundred steps.

Table 1: Oracle evidence rate of the loop test relative to the optimal rate ρ (bits/step), using the asymptotically most visited reference state ($\arg \max \pi$), which the burn-in rule selects with high probability. Simulation of up to 6×10^5 loops per chain; $\pm$ one standard error. The true ratio is at most 1; the entry above 1 is within Monte Carlo error.

Chain	ρ (bits/step)	Loop-test rate	Efficiency
ring p=0.3 q=0.25	0.0033	0.0033 ± 0.0001	1.002 ± 0.022
ring p=0.3 q=0.2	0.0145	0.0136 ± 0.0003	0.936 ± 0.020
random k=4 #0	0.0189	0.0174 ± 0.0004	0.920 ± 0.020
ring p=0.3 q=0.12	0.0575	0.0458 ± 0.0009	0.797 ± 0.015
random k=4 #1	0.0599	0.0480 ± 0.0010	0.801 ± 0.017
ring p=0.3 q=0.05	0.1429	0.0809 ± 0.0009	0.566 ± 0.006
random k=4 #2	0.1937	0.1116 ± 0.0015	0.576 ± 0.007
one-way ring p=0.3	0.3000	0.1014 ± 0.0006	0.338 ± 0.002

Why loops. Kolmogorov’s criterion states that a chain is reversible exactly when every cycle has equal forward and backward probability [31], that is, zero cycle affinity [14]. The loop test bets on this criterion along the cycles the trajectory actually traverses. It uses only the orientation of each loop, so, unlike the likelihood e-process, it does not pay to fit the symmetric part of T . We believe this is why its overhead is smaller (Section 7), although we have not proved it. A chain whose transition graph is a tree is always reversible, and on such a graph every loop factor is exactly 1.

7 Experiments

All experiments are seeded; `scripts/sequential_experiments.py` regenerates the ancillary files in `anc/`, and `scripts/sequential_tables.py` and `scripts/plot_sequential.py` regenerate the tables and figures. Throughout, $\alpha = 0.05$ and chains start from their stationary law.

Validity. Table 2 and Figure 1 report false-alarm rates over 1,000 trajectories of 20,000 steps for each reversible setting. The loop test is checked after every step. The likelihood e-process is checked at 303 checkpoints: every step up to 64, then a geometric grid. The likelihood-ratio test (LRT) is evaluated in three ways: once, at the planned $n = 20,000$; at 10 geometrically spaced looks from step 100; and at the 221 of those 303 checkpoints that come at or after step 100. The LRT uses the χ^2 reference distribution with $(k-1)(k-2)/2$ degrees of freedom, the number of independent cycle constraints on a complete graph.

Both anytime tests stay within level. The likelihood e-process raised no false alarms in 8,000 runs; it is conservative, as its overhead predicts. The LRT is calibrated at its planned sample size (4.2–5.7%) but rejects 23.9–35.5% of reversible trajectories when checked 10 times and 44.5–59.8% when checked at all 221 checkpoints. The birth–death chain’s transition graph is a tree, so every loop factor is exactly 1 (Section 6).

Three settings observe a reversible hidden chain through a noisy or lumping channel. The observed processes are reversible but not Markov, so no theorem above covers them. The loop test’s false-alarm rate there is 2.6–4.5%, within level.

Detection. Table 3 and Figure 2 report detection times over 400 trajectories per irreversible chain. The comparison is a fixed- n LRT with critical values calibrated by simulating 1,000 trajec-

Table 2: False-alarm rates on reversible processes at $\alpha = 0.05$ (1,000 trajectories of 20,000 steps; 95% Clopper–Pearson intervals for the loop test). “Hidden” rows are reversible but not Markov.

Reversible process	Type	Loop test	Likelihood	LRT, fixed n	LRT, 10 looks	LRT, 221 checkpoints
random $k=3$	Markov	3.7% [2.6, 5.1]	0.0%	5.7%	30.1%	53.2%
random $k=4$	Markov	3.5% [2.4, 4.8]	0.0%	4.4%	30.4%	51.1%
random $k=5$	Markov	3.1% [2.1, 4.4]	0.0%	5.5%	35.5%	59.8%
birth-death $k=5$	Markov	0.0% [0.0, 0.4]	0.0%	n/a	n/a	n/a
unbiased ring $k=3$	Markov	2.6% [1.7, 3.8]	0.0%	5.5%	29.1%	49.4%
reversible $k=4$, noisy view	hidden	4.5% [3.3, 6.0]	0.0%	4.7%	28.7%	47.9%
reversible $k=4$, states 2,3 merged	hidden	2.6% [1.7, 3.8]	0.0%	4.7%	28.6%	48.3%
unbiased ring, noisy view	hidden	3.1% [2.1, 4.4]	0.0%	4.2%	23.9%	44.5%

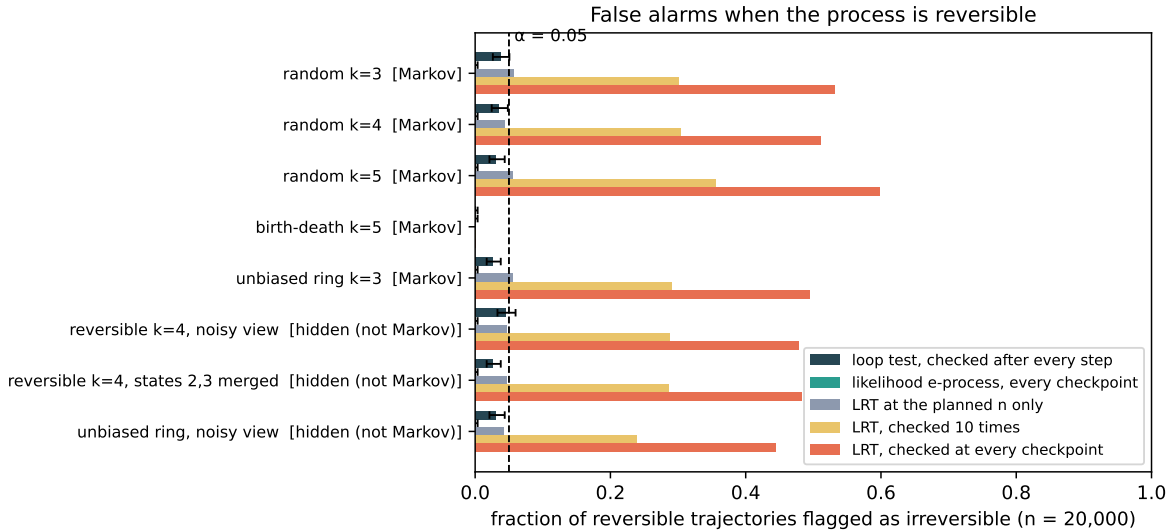


Figure 1: False alarms on reversible processes. The loop test is checked after every step; the likelihood e-process at 303 checkpoints; the LRT once, at 10 looks, or at 221 checkpoints from step 100.

tories of the least favorable reversible chain M . Its sample size is the smallest n with 80% power against the true chain. This baseline is idealized: it is told the right sample size in advance, which no experimenter knows. We calibrate its cutoffs by simulation because the asymptotic χ^2 cutoffs are unreliable at small n . With the asymptotic cutoffs the baseline would look cheaper for seven of the eight chains, by up to 36% for the chains far from equilibrium, where the asymptotic cutoffs are anti-conservative. For one chain (random $k = 4$ #1) they are conservative instead.

For the five chains with $\rho < 0.07$ bits/step, the loop test’s 80th-percentile detection time is 1.7–1.9 times the fixed-sample size, and its median roughly equals that size. Far from equilibrium the ratio rises to 2.5–4.2, because of the burn-in and the discarded within-loop information. These chains are detected within about a hundred steps. The likelihood e-process needs 3.5–8.7 times the fixed-sample size. No median falls below the floor $\log_2(20)/\rho$ of Theorem 4.3. Medians and percentiles carry Monte Carlo error of about $\pm 8\%$.

Growth rates. Over 20 trajectories of 2×10^5 steps, the likelihood e-process’s $\frac{1}{t} \log_2 E_t$ converges to ρ (Figure 3). The final rates and ρ are 0.1430 vs 0.1429 and 0.0143 vs 0.0145 for rings with $q = 0.05$ and $q = 0.2$, 0.2997 vs 0.3000 for a one-way ring with $\sigma = \infty$, and 0.1605 vs 0.1614 for a

Table 3: Detection times (steps) on irreversible chains at $\alpha = 0.05$: the fixed- n LRT sample size for 80% power (calibrated by simulation), the loop test’s median and 80th percentile, and the likelihood e-process’s 80th percentile. Ratios are relative to the fixed- n LRT. The floor $\log_2(20)/\rho$ is the asymptotic bound of Theorem 4.3.

Chain	ρ	$\log_2(20)/\rho$	Fixed- n LRT	Loop test (median / 80%)	Likelihood (80%)
ring $p=0.3$ $q=0.25$	0.0033	1316	1818	1907 / 3308 (1.8 $\times$)	15686 (8.6 $\times$)
ring $p=0.3$ $q=0.2$	0.0145	298	417	407 / 701 (1.7 $\times$)	2834 (6.8 $\times$)
random $k=4$ #0	0.0189	229	405	436 / 789 (1.9 $\times$)	3539 (8.7 $\times$)
ring $p=0.3$ $q=0.12$	0.0575	75	101	123 / 190 (1.9 $\times$)	557 (5.5 $\times$)
random $k=4$ #1	0.0599	72	119	146 / 226 (1.9 $\times$)	913 (7.7 $\times$)
ring $p=0.3$ $q=0.05$	0.1429	30	37	91 / 110 (3.0 $\times$)	184 (5.0 $\times$)
random $k=4$ #2	0.1937	22	39	83 / 99 (2.5 $\times$)	198 (5.1 $\times$)
one-way ring $p=0.3$	0.3000	14	22	82 / 93 (4.2 $\times$)	77 (3.5 $\times$)

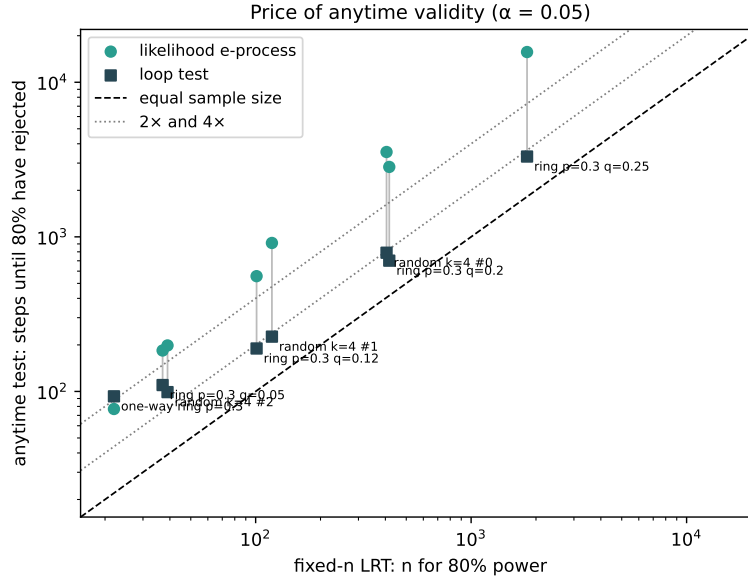


Figure 2: Price of anytime validity: steps until 80% of trajectories have rejected, against the idealized fixed-sample LRT sample size.

fourth random four-state chain (distinct from those in Table 3). The loop test settles at its lower oracle rate.

ρ versus σ . Over 300 random chains, ρ lies below $\sigma/2$ and tracks $\sigma/4$ near equilibrium, bending further below it as σ grows (Figure 4).

AI-assisted review and reproduction. As a consistency check, and not as independent peer review, we used separate AI-agent sessions without access to our working context.

- *Adversarial review of the proofs and code.* It found a gap in an earlier version, which averaged loop tests over reference states; that version was replaced. It exhibited a step-level stopping rule with $\mathbb{E}[E_\tau] \approx 1.18$ for the loop test (Section 6), and it checked the reversible maximum-likelihood iteration against multi-start quasi-Newton optimization on 150 random count matrices

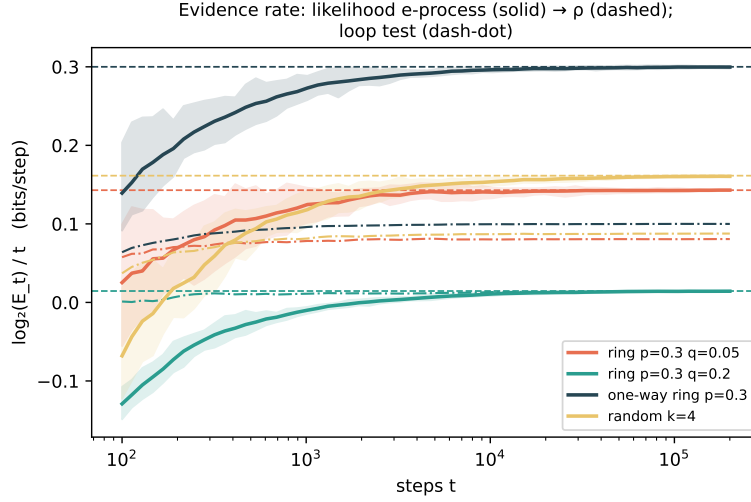


Figure 3: Evidence growth. Solid: likelihood e-process (10–90% band); dashed: ρ ; dash-dot: loop test.

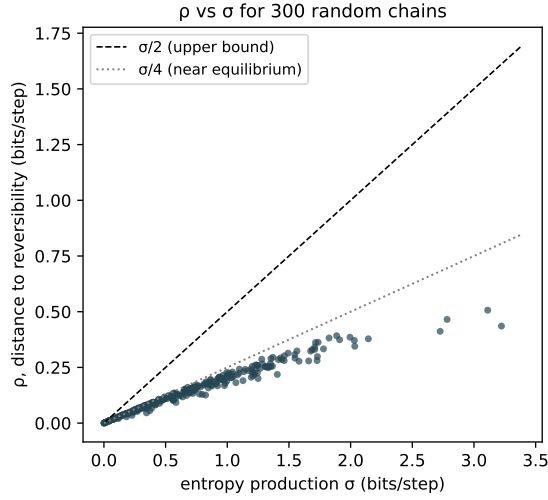


Figure 4: The optimal detection rate ρ against entropy production σ for 300 random chains.

(Section 5). It also found that the baseline LRT was anti-conservative; the LRT is now calibrated by simulation.

- *Reproduction.* A second session regenerated the growth, efficiency, and rate experiments with byte-identical output. It also reimplemented the loop test from the written specification alone. It obtained 3.8% [2.5, 5.7] false alarms on a reversible four-state chain, and a median detection time of 394 steps against our 407 for the $q = 0.2$ ring.

These checks reduce, but do not replace, the need for human review.

8 Discussion

Summary. Deciding whether a system is at equilibrium, from data examined as they arrive, is governed by ρ , the divergence to the nearest reversible chain, rather than by the entropy production σ . The loop test makes this decision with an anytime-valid false-alarm guarantee whose proof is elementary. For $\rho < 0.07$ bits/step, its cost is 1.7–1.9 times an idealized fixed-sample design.

Limitations.

- The theory covers fully observed, finite-state, first-order Markov chains. For hidden-state records the tests stayed within level in our simulations, but the guarantees do not apply. Excursions of an observed state need not regenerate the hidden state.
- The tests certify that detailed balance is broken; they do not give a confidence bound on σ . Because $\sigma \geq 2\rho$, a time-uniform lower confidence bound on ρ would bound σ from below; constructing one is open.
- Far from equilibrium the loop test discards within-loop information: its efficiency falls to 0.57 at $\rho \approx 0.14$ and 0.34 for the one-way ring. Combining it with the likelihood e-process would require a common filtration, which Theorem 6.1’s loop-indexed martingale does not provide.
- Efficiency depends on the reference state. We report the most visited state, which the burn-in rule selects with high probability; we did not study other choices.
- The experiments use small state spaces ($k \leq 5$) and synthetic chains.

Extensions. Natural next steps are loop tests for hidden Markov models through regeneration states, higher-order and continuous-time versions, and time-uniform confidence sequences for ρ .

Code and data

Code, ancillary data, and scripts are at <https://github.com/jbwinters/Evidence-Holonomy>. `scripts/reproduce_all.sh` regenerates every number, table, and figure, and this document.

Acknowledgements

Portions of the text, code, and analysis were drafted or revised with assistance from AI language models (OpenAI’s GPT-5 and Anthropic’s Claude), including the AI-agent review and reproduction described in Section 7. The author verified the content and takes full responsibility for the paper.

A Estimating entropy production

The earlier version of this work studied an estimator of σ itself. We summarize it here as a companion to the tests.

The plug-in estimator. Given $x_{0:n-1}$, let $y = x_{1:n-1}$ and $\tilde{y}$ its reversal. Fit KT mixtures over Markov orders $0, \dots, R$ to y and to $\tilde{y}$, freeze them as predictors $\hat{p}$ and $\hat{q}$, and set

$$\hat{\sigma}_{n,R} = \frac{1}{n-1} \sum_t \log_2 \frac{\hat{p}(y_t \mid y_{t-R:t-1})}{\hat{q}(y_t \mid y_{t-R:t-1})}.$$

This is an in-sample plug-in estimate of the divergence of [18, 19]. For an irreducible chain with $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$ and $R \geq 1$, $\hat{\sigma}_{n,R} \rightarrow \sigma$ almost surely. Block frequencies of y and $\tilde{y}$ converge to those of P and P^* , the mixture posterior abandons the inadequate order 0, and the ergodic theorem applies. Estimates for different $R \geq 1$ nearly coincide. That reflects a shared, correctly specified model class, not independence from the choice of coder.

Why universality alone does not justify KL estimation. Pointwise universality of a code controls code lengths only on paths typical for the law it is universal for. Distinct stationary ergodic laws are mutually singular, so a universal code for P^* says nothing about the code length of a path drawn from P . Differences of separate compressions estimate the entropy-rate difference, which vanishes for time reversal. Estimating $D(P\|P^*)$ requires a model of P^* that is accurate on contexts typical for P , and a finite-order plug-in supplies one for finite-state Markov chains.

Hidden-state records. If Y observes a Markov chain Z through a memoryless channel, then reversing the observation equals observing the reversal. By data processing, $D_m := D(P_m^Y\|P_m^{Y*}) \leq (m-1)\sigma_Z$ for every m [7]. The order- R estimator targets the conditional increment $d_R = D_{R+1} - D_R$. Averages of these increments are bounded by σ_Z , but a single d_R need not be.

Numerical validation. Table 4 and Figure 5 follow 15 random chains (σ from 0.0067 to 1.92 bits/step). The median relative error falls from 10.3% at $n = 2^9$ to 1.3% at $n = 2^{15}$. On 60 further chains at $n = 2^{15}$, the correlation with σ is 0.9994 (Figure 6). On reversible chains the positive in-sample bias shrinks roughly as $1/n$: 0.0015 bits/step at $n = 2^{11}$ and 0.00013 at 2^{15} for three states. A hidden three-state ring with $\sigma_Z = 0.646$ seen through a noisy channel yields 0.085–0.127 bits/step. Merging two of its states makes the observed process exactly reversible, and the estimate falls to at most 7×10^{-5} (Figure 7).

Table 4: Absolute relative error $|\hat{\sigma} - \sigma|/\sigma$ of the time-reversal estimate ($R = 3$) on 15 random biased chains (five each with 3, 4, and 5 states), by sample length n . Every row uses the same 15 chains.

n	median	interquartile range	max
2^9	10.3%	[2.2%, 14.1%]	123%
2^{11}	8.8%	[5.3%, 11.8%]	86%
2^{13}	4.4%	[3.7%, 7.7%]	31%
2^{15}	1.3%	[0.3%, 3.9%]	19%

B Evidence holonomy

The earlier version of this work framed irreversibility as a “holonomy” of evidence: the change in description length when a path is carried around a loop of representation transforms and scored either in the transformed representation or under the loop’s pushforward law. For time reversal, the first version vanishes and the second equals σ for Markov chains [6]. Every other loop we examined reduced to a bijection, which gives zero, or to a deterministic coarse-grain-and-lift, which gives an infinite rate. The general formalism therefore organizes known results but does not produce new ones, and we do not pursue it here.

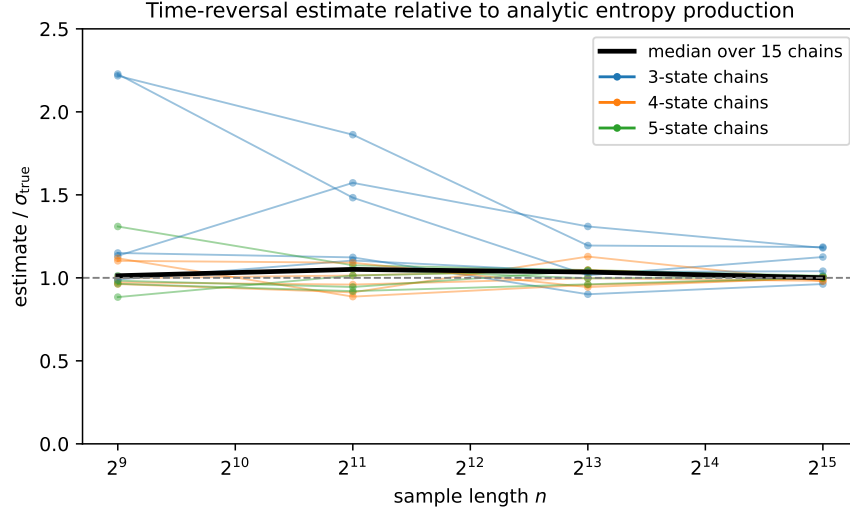


Figure 5: Estimate divided by σ for 15 random chains and their median, as the sample length grows.

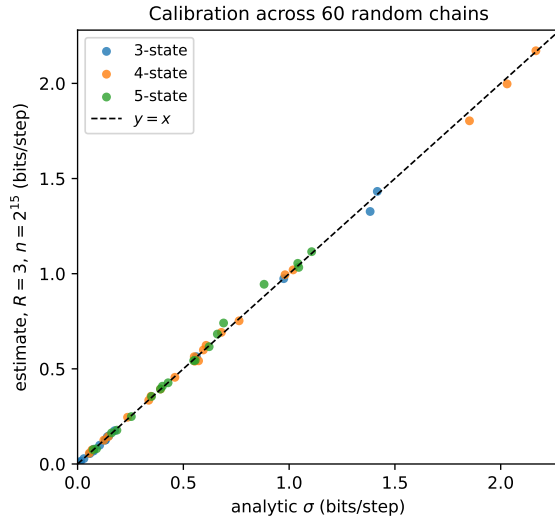


Figure 6: Estimate ($R = 3$, $n = 2^{15}$) against σ for 60 random chains.

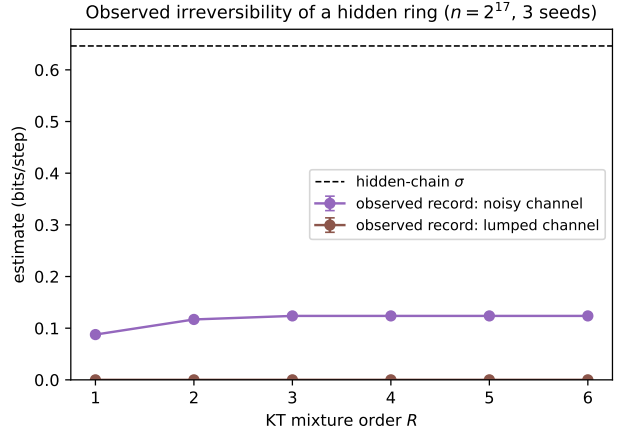


Figure 7: Estimates on observed records of a hidden ring, by mixture order.

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